

Calculus – Real sequences

Family 1. The numerator and the denominator is a polynomial expression of n . In that case convergence can be investigated by the degrees of the polynomials. Denote the polynomial in the numerator and in the denominator by p and q , respectively.

- (1) If $\deg(p) < \deg(q)$, then the sequence converges and tends to 0 as $n \rightarrow \infty$.
- (2) If $\deg(p) = \deg(q)$, then the sequence converges and tends to the ratio of leading coefficients as $n \rightarrow \infty$.
- (3) If $\deg(p) > \deg(q)$, then the sequence diverges (it is not convergent).

Example 1. Investigate the sequence

$$a_n := \frac{(2n+1)^3 + n^2 + 1}{n^3 + 2n + 10}$$

for convergence.

Solution. In this case $p(n) = (2n+1)^3 + n^2 + 1$ and $q(n) = n^3 + 2n + 10$. Here $(2n+1)^3 = 8n^3 + 6n^2 + 6n + 1$, hence $\deg(p) = 3$. For q this is obviously simpler, that is, $\deg(q) = 3$. Thus we are in case (2), and hence (a_n) is convergent.

The limit in this case is the ratio of the leading coefficients (or main coefficients) of p and q , which are 8 and 1, respectively. Consequently,

$$\lim_{n \rightarrow \infty} \frac{(2n+1)^3 + n^2 + 1}{n^3 + 2n + 10} = 8.$$

We note that since the sequence is convergent, it is also bounded.

Example 2. Investigate the sequence

$$a_n := \frac{20n + \pi}{(n+1)^2 + 3}$$

for convergence.

Solution. Obviously, we have $\deg(p) = 1$ and $\deg(q) = 2$, where $p(n) = 20n + \pi$ and $q(n) = (n+1)^2 + 3$. Hence this sequence belong to case (1), consequently it is convergent and $a_n \rightarrow 0$ as $n \rightarrow \infty$ (i.e., it is a null sequence).

Example 3. Investigate the sequence

$$a_n := \frac{2n - 7n^6 + 13}{(2n^2 + 1)^2 + 8}$$

for convergence.

Solution. Let $p(n) := 2n - 7n^6 + 13$ and $q(n) := (2n^2 + 1)^2 + 8$. Then, after applying the identity $(a+b)^2 = a^2 + 2ab + b^2$, we obtain that

$$q(n) = (2n^2 + 1)^2 + 8 = 4n^4 + 4n^2 + 1 + 8 = 4n^4 + 4n^2 + 9.$$

Hence $\deg(p) = 6$ and $\deg(q) = 4$, which yields that this sequence belongs to case (3). Accordingly, (a_n) diverges.

More precisely, the sequence (a_n) is convergent in the extended sense. To determine its limit, we reformulate the expression as follows.

$$\frac{2n - 7n^6 + 13}{(2n^2 + 1)^2 + 8} = \frac{n^6 \cdot \left(\frac{2}{n^5} - 7 + \frac{13}{n^6}\right)}{n^4 \cdot \left(4 + \frac{4}{n^2} + \frac{9}{n^4}\right)} = n^2 \cdot \frac{-7 + \frac{2}{n^5} + \frac{13}{n^6}}{4 + \frac{4}{n^2} + \frac{9}{n^4}}.$$

Here $n^2 \rightarrow +\infty$ as $n \rightarrow \infty$ and

$$\frac{-7 + \frac{2}{n^5} + \frac{13}{n^6}}{4 + \frac{4}{n^2} + \frac{9}{n^4}} \rightarrow -\frac{7}{4} \quad \text{as } n \rightarrow \infty.$$

Consequently, $a_n \rightarrow -\infty$ as $n \rightarrow \infty$. (The calculation can be shortened by the following observation: Based on the degrees, the sequence diverges either to $+\infty$ or $-\infty$. But the sign of the ratio of the leading coefficients is negative. Hence the limit is $-\infty$.)

Family 2. The numerator and denominator are sums of products, where one factor of the product is a polynomial and the other is an exponential function. In that case we can apply the following theorem.

• Let $r \in \mathbb{R}$ with $-1 < r < 1$ and $p : \mathbb{N} \rightarrow \mathbb{R}$ be a polynomial. Then the sequence $(p(n) \cdot r^n)$ is convergent and tends to 0 as $n \rightarrow \infty$. (Note that the condition on the basis of the exponential function is essential. If the basis is less than or equal to -1 , or greater than or equal to 1 , the theorem does not hold.)

Example 4. Investigate the sequence

$$a_n := \frac{(n+1)^2 \cdot 3^{n-2} + (2n+1) \cdot 2^n}{(-5)^{n+1} - n^3 \cdot 4^{n-1}}$$

for convergence.

Solution. First we determine the basis with the largest absolute value.

$$\max(|3|, |2|, |-5|, |4|) = \max(2, 3, 4, 5) = 5.$$

Hence we simplify a_n by 5^n .

$$a_n := \frac{(n+1)^2 \cdot 3^{n-2} + (2n+1) \cdot 2^n}{(-5)^{n+1} - n^3 \cdot 4^{n-1}} = \frac{\frac{1}{9}(n+1)^2 \cdot \left(\frac{3}{5}\right)^n + (2n+1) \cdot \left(\frac{2}{5}\right)^n}{(-5) \cdot (-1)^n - \frac{1}{4}n^3 \cdot \left(\frac{4}{5}\right)^n}.$$

Here the blocks

$$\frac{1}{9}(n+1)^2 \cdot \left(\frac{3}{5}\right)^n, \quad (2n+1) \cdot \left(\frac{2}{5}\right)^n, \quad \text{and} \quad \frac{1}{4}n^3 \cdot \left(\frac{4}{5}\right)^n$$

tend to 0 as $n \rightarrow \infty$ (because of the theorem above!). The only sequence in our expression that is divergent is $(-1)^n$. For further computation, we introduce some notation to shorten the expressions and decompose the original sequence into even and odd parts (to handle $(-1)^n$).

Let

$$b_n := \frac{1}{9}(n+1)^2 \cdot \left(\frac{3}{5}\right)^n, \quad c_n := (2n+1) \cdot \left(\frac{2}{5}\right)^n, \quad \text{and} \quad d_n := \frac{1}{4}n^3 \cdot \left(\frac{4}{5}\right)^n.$$

As stated earlier, these are null sequences, so any subsequence of them is convergent and tends to 0. Consequently, the even and odd parts of these sequences are also null sequences. Using these, we obtain that

$$a_{2n} = \frac{b_{2n} + c_{2n}}{(-5) \cdot (-1)^{2n} - d_{2n}} = \frac{b_{2n} + c_{2n}}{-5 - d_{2n}} \rightarrow \frac{0+0}{-5+0} = 0 \quad \text{as } n \rightarrow \infty$$

and

$$a_{2n-1} = \frac{b_{2n-1} + c_{2n-1}}{(-5) \cdot (-1)^{2n-1} - d_{2n-1}} = \frac{b_{2n-1} + c_{2n-1}}{5 - d_{2n-1}} \rightarrow \frac{0+0}{5+0} = 0 \quad \text{as } n \rightarrow \infty.$$

We can see that the sequences (a_{2n}) and (a_{2n-1}) have the same limit, which implies that (a_n) is convergent and tends to the common limit of the previous sequences, that is, $a_n \rightarrow 0$ as $n \rightarrow \infty$.

Example 5. Investigate the sequence

$$a_n := \frac{(n^3 + 2) \cdot 3^{2n+1} + n^2 \cdot 5^{n+1}}{2^n + (3n^3 + 6) \cdot 9^n}$$

for convergence.

Solution. The basis with maximal modulus is 9, hence we simplify with 9^n . This results

$$a_n = \frac{3(n^3 + 2) + 5n^2 \cdot \left(\frac{5}{9}\right)^n}{\left(\frac{2}{9}\right)^n + 3n^3 + 6}.$$

Then $5n^2 \cdot \left(\frac{5}{9}\right)^n \rightarrow 0$ and $\left(\frac{2}{9}\right)^n \rightarrow 0$ as $n \rightarrow \infty$. The remaining terms are divergent, since they are polynomials. To handle these, we simplify by the largest exponent of n . Then

$$\frac{3(n^3 + 2) + 5n^2 \cdot \left(\frac{5}{9}\right)^n}{\left(\frac{2}{9}\right)^n + 3n^3 + 6} = \frac{3 + \frac{6}{n^3} + \frac{5}{n} \cdot \left(\frac{5}{9}\right)^n}{\frac{1}{n^3} \cdot \left(\frac{2}{9}\right)^n + 1 + \frac{6}{n^3}} \rightarrow \frac{3+0+0}{0+1+0} = 3 \quad \text{as } n \rightarrow \infty.$$

We note that in the cases $\frac{5}{n} \cdot \left(\frac{5}{9}\right)^n \rightarrow 0$ and $\frac{1}{n^3} \cdot \left(\frac{2}{9}\right)^n \rightarrow 0$, instead of the theorem above, we used the fact that a product of a bounded sequence and a null sequence is a null sequence.